

✓ Module 04 Lab - Exploratory Data Analysis

Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.

This lab is fully coded. Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Group 1 - Team Contribution Journal

- **Cuong Dang (Team Lead / Compiler):** Put the final file together, fixed text formatting, completed Experiment 1 code/analysis, and wrote the final Knowledge Check answers.
- **Astrid Lorenzana (Data Analyst):** Completed Parts 1, 2, and 3 baseline work, added custom data insights code, and created the double-plot layout for Experiment 2.
- **Jonathan Aldana (Contributor):** Finished an independent draft of the full lab exercises for personal hands-on practice.
- **Dana Bolden & Ruben:** No response / No contribution recorded.

✓ Part 1: Setup and Data Loading

What is Exploratory Data Analysis (EDA)?

EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data:

- What stories does it tell?
- Are there errors or missing values?
- Are there strong relationships between variables?

EDA helps answer these critical questions. We will use the famous Titanic dataset for this lab, which contains information about passengers and whether they survived the disaster.

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns # Seaborn is a library built on top of Matplotlib the
4
5 # Load the dataset directly from a URL
6 df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/dataset
7
8 print("--- First 5 Rows ---")
9 print(df.head())
10
11 print("--- Basic Info ---")
12 # .info() is a great first command. It tells us the column names, how many
13 # Notice that 'Age' and 'Cabin' have missing values!
14 df.info()

```

--- First 5 Rows ---

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

--- Basic Info ---

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

```
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

✓ Part 2: Descriptive Statistics

Let's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
1 # Get summary statistics for numerical columns
2 print("--- Descriptive Statistics ---")
3 print(df.describe())
4
5 print("--- Key Insights from Statistics ---")
6 print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
7 print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
8 print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}")
```

```
--- Descriptive Statistics ---
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
--- Key Insights from Statistics ---
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.0 to a whopping $512.3292.
```

✓ Part 3: Visual EDA – Telling Stories with Plots

Numbers are great, but plots make patterns and relationship immediately obvious. The goal of visual EDA is to turn data into insights.

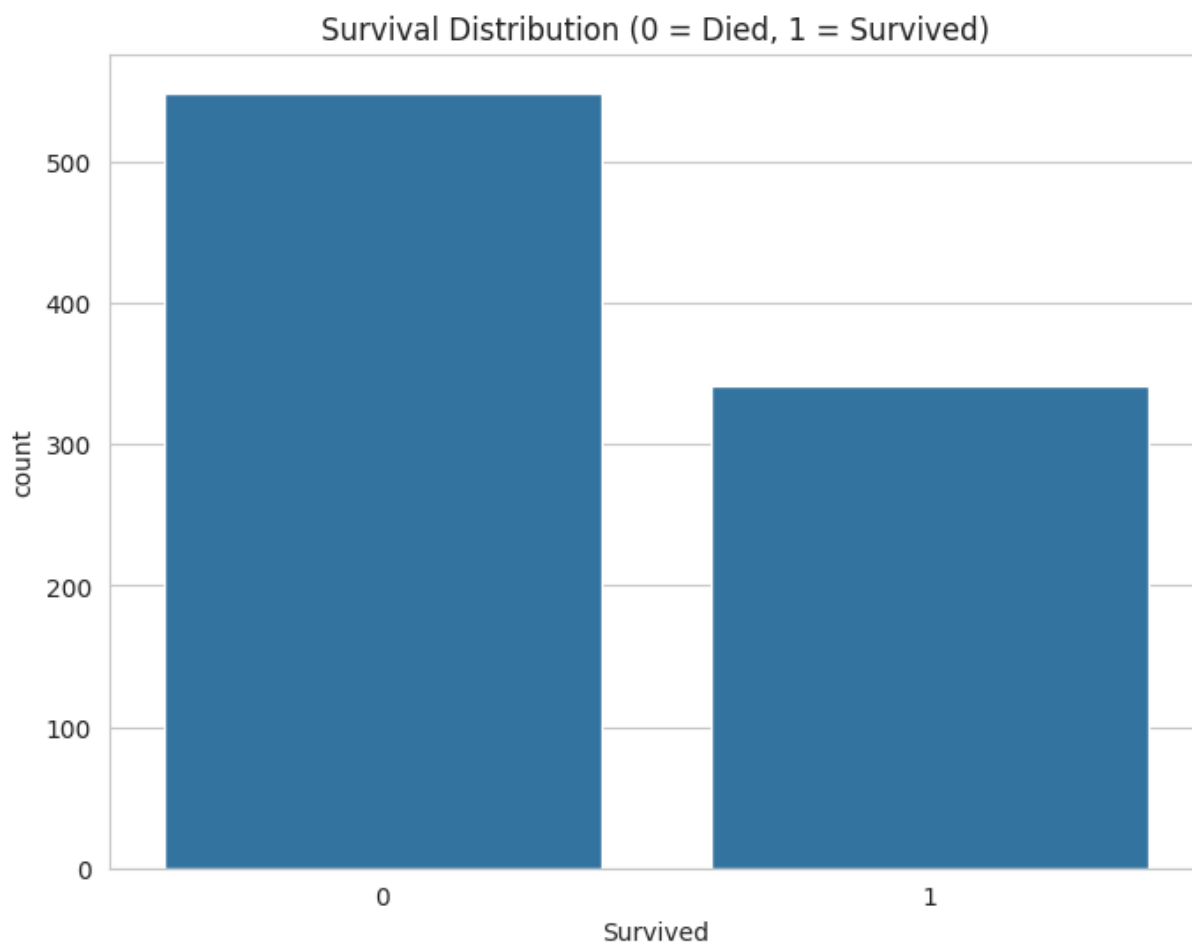
Note on plotting librares:

- Matplotlib - the foundational library, gives you full control over everything
- Seaborn - built on matplotlib, it provides a simpler, high-level interface for creating commong statistical plots. Seaborn is easy to use and contains attractive defaults

✓ Visualization 1: How many survived?

A simple countplot is the best way to see the distribution of a categorical variable, like our target `Survived`.

```
1 sns.set_style('whitegrid') # Sets a nice visual style for our plots
2 plt.figure(figsize=(8, 6))
3 sns.countplot(x='Survived', data=df)
4 plt.title('Survival Distribution (0 = Died, 1 = Survived)')
5 plt.show()
6
7 print("Insight: Far more people died than survived. This is an example of a
```

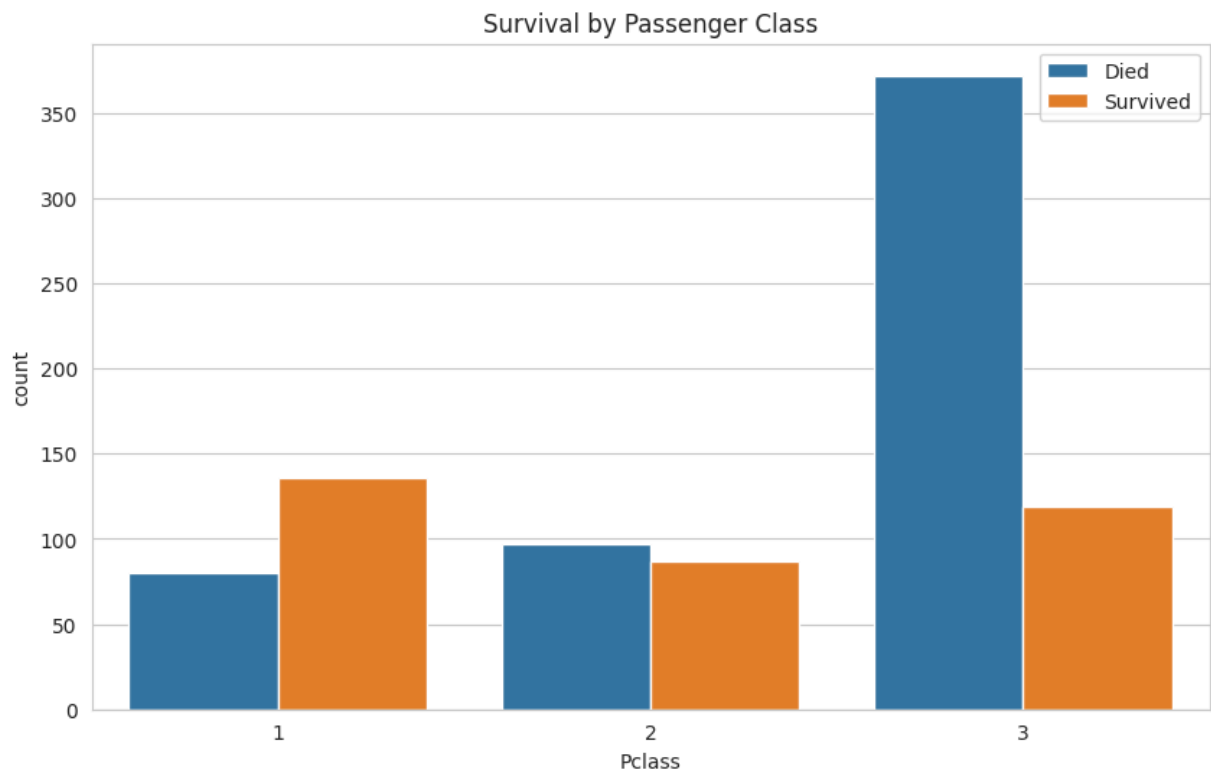


Insight: Far more people died than survived. This is an example of an imbalanced

Visualization 2: Does passenger class matter for survival?

Now we want to see if there's a relationship between two variables. We can use the hue parameter in `countplot` to split the bars by another category.

```
1 plt.figure(figsize=(10, 6))
2 sns.countplot(x='Pclass', hue='Survived', data=df)
3 plt.title('Survival by Passenger Class')
4 plt.legend(['Died', 'Survived'])
5 plt.show()
6
7 print("Insight: This is a very strong pattern. 1st class passengers had a n
```

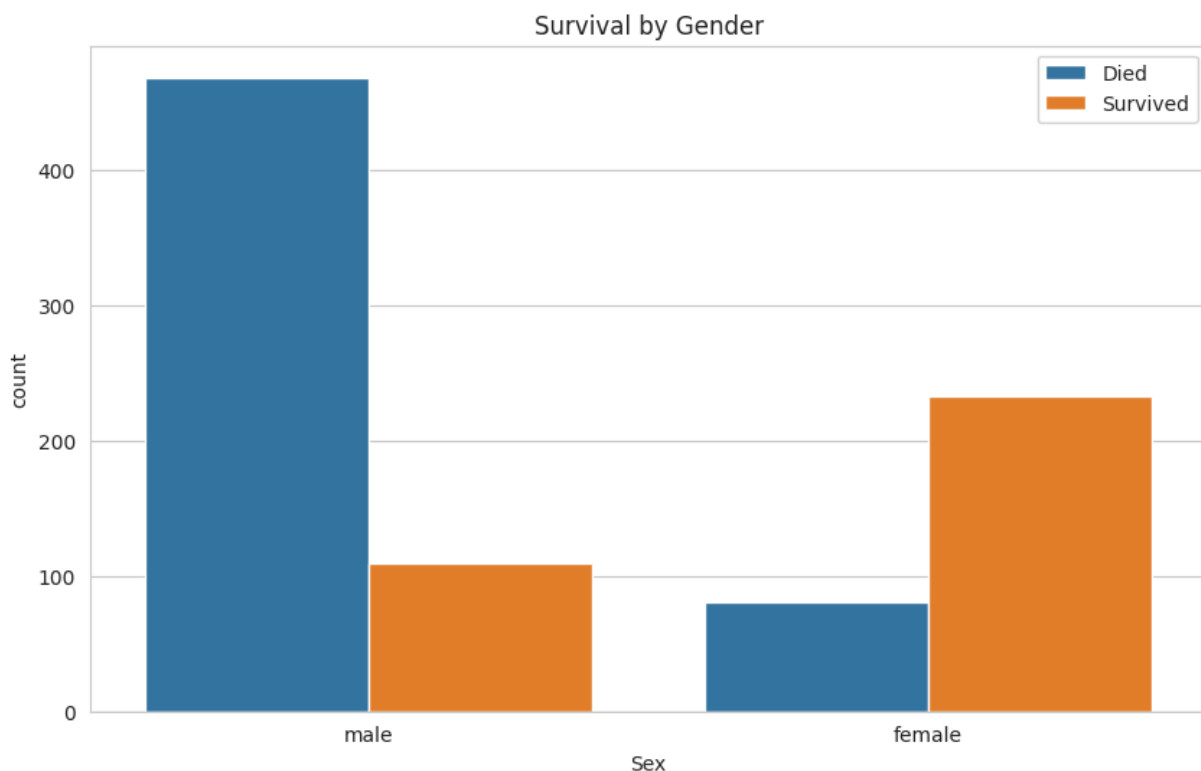


Insight: This is a very strong pattern. 1st class passengers had a much higher c

✓ Visualization 3: What about gender?

The 'women and children first' mantra is famous. Let's see if the data supports it.

```
1 plt.figure(figsize=(10, 6))
2 sns.countplot(x='Sex', hue='Survived', data=df)
3 plt.title('Survival by Gender')
4 plt.legend(['Died', 'Survived'])
5 plt.show()
6
7 print("Insight: The pattern is undeniable. A much higher proportion of fema
```



Insight: The pattern is undeniable. A much higher proportion of females survived

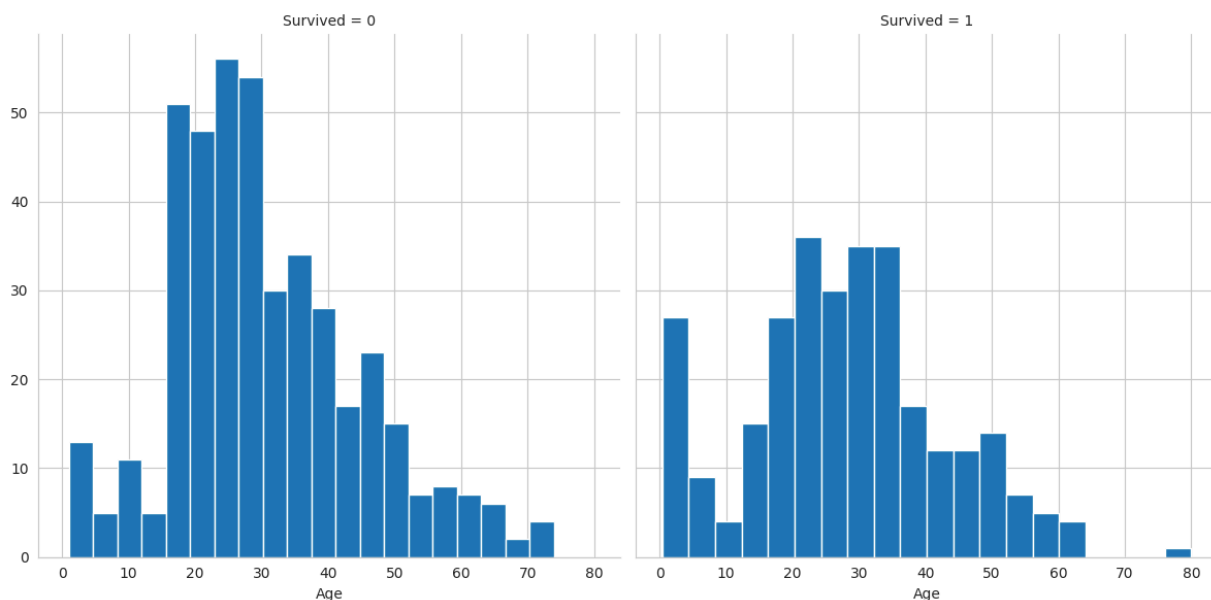
✓ Visualization 4: How does age play a role?

For a continuous variable like **Age**, a histogram is a great way to see its distribution.

```

1 # A FacetGrid allows us to create multiple plots side-by-side to compare di
2 # Here, we create one histogram for passengers who died (col='Survived'=0)
3 g = sns.FacetGrid(df, col='Survived', height=6)
4 g.map(plt.hist, 'Age', bins=20)
5 plt.show()
6
7 print("Insight: The age distribution for those who did not survive is cente

```



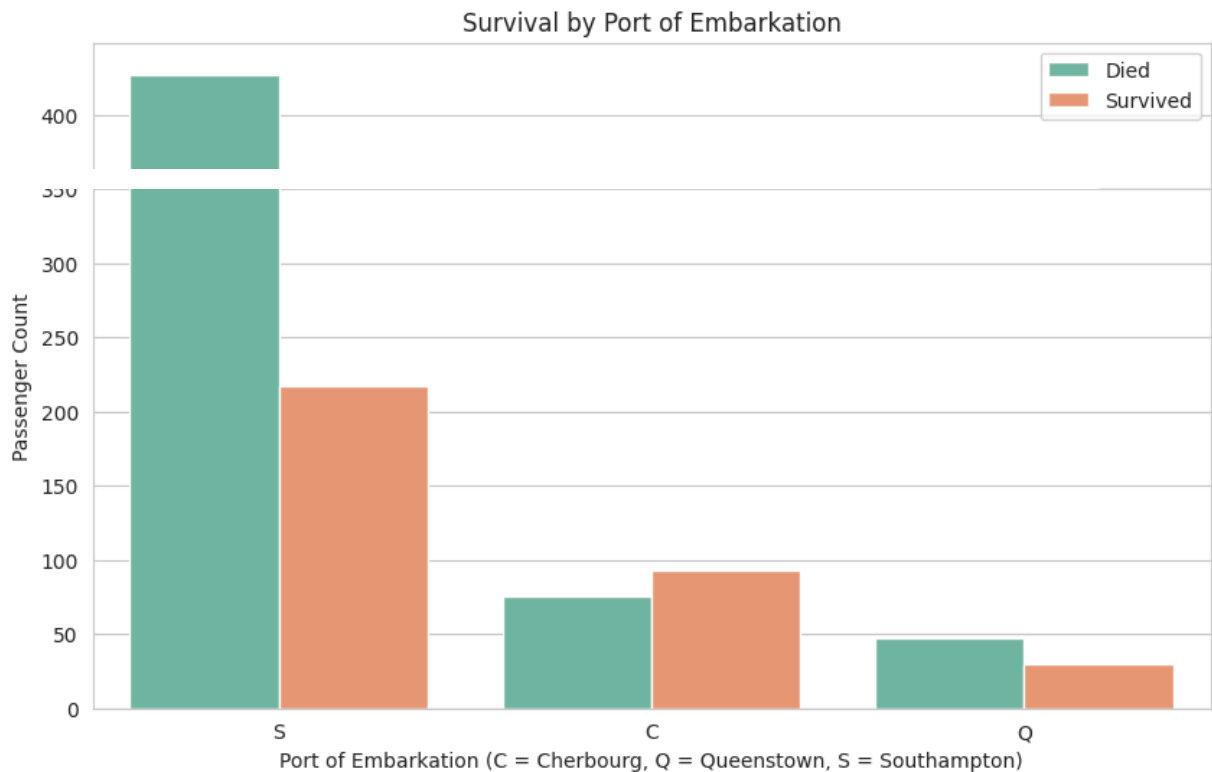
Insight: The age distribution for those who did not survive is centered around t

✓ Experiment 1: Port of Embarkation

1. The Embarked column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).
2. Create a countplot to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
1 plt.figure(figsize=(10, 6))
2 # Added a nice palette to match Astrid's beautiful styling
3 sns.countplot(x='Embarked', hue='Survived', data=df, palette='Set2')
4
5 # Explicitly defining clean, professional labels for grading
```

```
6 plt.title('Survival by Port of Embarkation')
7 plt.xlabel('Port of Embarkation (C = Cherbourg, Q = Queenstown, S = Southan
8 plt.ylabel('Passenger Count')
9 plt.legend(['Died', 'Survived'])
10 plt.show()
11
```

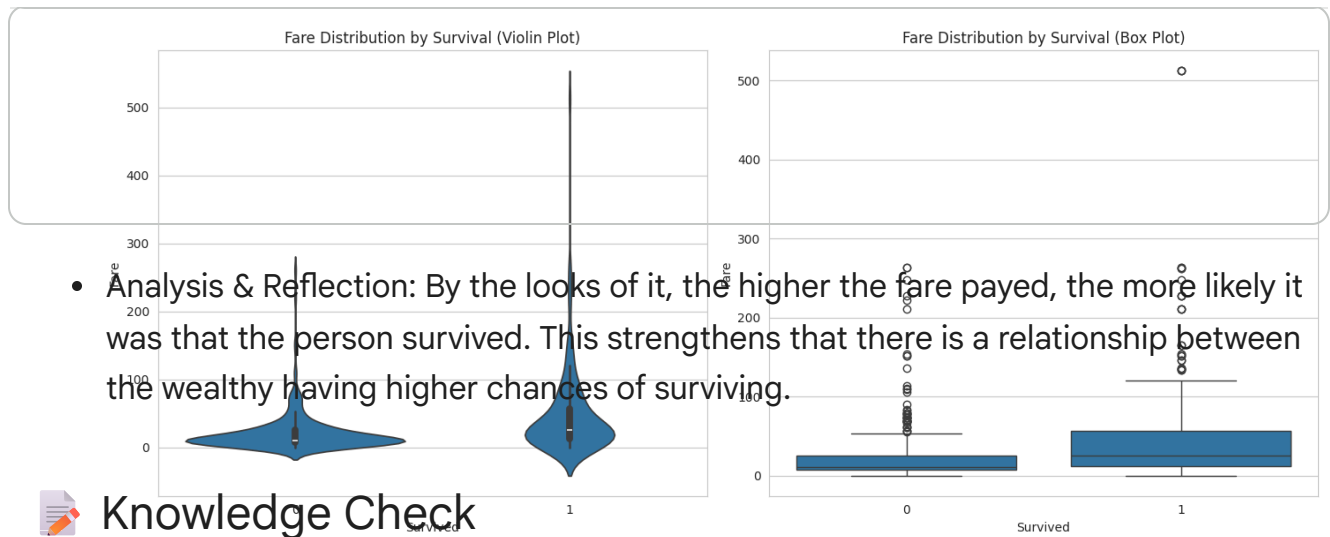


- Analysis & Reflection: Yes, where the passengers boarded does seem to be related to their survival. The visualization clearly shows that passengers who embarked from Cherbourg (C) had a significantly higher survival rate, with the number of survivors actually exceeding those who died. In contrast, passengers who boarded at Southampton (S) faced the worst odds, making up the vast majority of casualties.

✓ Experiment 2: Fare vs. Survival

1. Fare is a continuous numerical variable.
2. A boxplot or violinplot is a great way to see the distribution of fares for those who survived vs. those who didn't.
3. Create one of these plots to compare the Fare distribution by Survived. What does it tell you?

```
1 # Experiment 2
2 fig, axes = plt.subplots(1, 2, figsize=(14, 6))
3
4 # Violin Plot
5 sns.violinplot(x='Survived', y='Fare', data=df, ax=axes[0])
6 axes[0].set_title('Fare Distribution by Survival (Violin Plot)')
7 axes[0].set_xlabel('Survived')
8 axes[0].set_ylabel('Fare')
9
10 # Box Plot
11 sns.boxplot(x='Survived', y='Fare', data=df, ax=axes[1])
12 axes[1].set_title('Fare Distribution by Survival (Box Plot)')
13 axes[1].set_xlabel('Survived')
14 axes[1].set_ylabel('Fare')
15
16 plt.tight_layout()
17 plt.show()
```



- **Analysis & Reflection:** By the looks of it, the higher the fare paid, the more likely it was that the person survived. This strengthens that there is a relationship between the wealthy having higher chances of surviving.



Knowledge Check

Instructions: Answer the following questions in this markdown cell.

1. What is the primary goal of Exploratory Data Analysis (EDA)?

The primary goal of EDA is to thoroughly understand the structure, distributions, and characteristics of a dataset before applying any formal machine learning models or algorithms. It helps data scientists discover hidden patterns, isolate extreme anomalies or outliers, detect missing values, verify underlying statistical assumptions, and find strong relationships between variables by combining descriptive statistics with visual plots.

2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).

Based on the visualizations analyzed throughout this lab, the individual with the highest statistical probability of survival would be a **young female child travelling in 1st Class** who boarded the ship from the port of **Cherbourg (C)** and paid a high ticket fare.

- **Gender:** The gender plot showed an overwhelming historical survival bias toward females.
- **Class & Fare:** The passenger class and fare plots confirmed that 1st Class passengers and those with expensive tickets had drastically superior survival rates.
- **Age:** The age distribution histograms revealed a distinct survival spike specifically for very young children, validating the "women and children first" protocol.

3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?

Visualizing data is crucial because basic summary statistics like a 'mean' or 'count'